

Aceso: PICO-Guided Evidence Summarization on Medical Literature

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Abstract—Evidence-Based Medicine (EBM) aims to apply the best available evidence gained from scientific methods to clinical decision making. A generally accepted criterion to formulate evidence is to use the PICO framework, where PICO stands for Problem/Population, Intervention, Comparison, and Outcome. Automatic extraction of PICO-related sentences from medical literature is crucial to the success of many EBM applications. In this work, we present our Aceso¹ system, which automatically generates PICO-based evidence summaries from medical literature. In Aceso, we adopt an active learning paradigm, which helps to minimize the cost of manual labeling and to optimize the quality of summarization with limited labeled data. An UMLS2Vec model is proposed to learn a vector representation of medical concepts in UMLS,² and we fuse the embedding of medical knowledge with textual features in summarization. The evaluation shows that our approach is better on identifying PICO sentences against state-of-the-art studies and outperforms baseline methods on producing high-quality evidence summaries.

Index Terms—Evidence-Based Medicine, PICO, Summarization, Active Learning.

I. INTRODUCTION

EVIDENCE-BASED Medicine (EBM) [1] emphasizes the use of evidence from well-designed and well-conducted research to provide clinical decision support. In EBM, the practical use of evidence usually depends on manually searching and analyzing over a large medical literature database such as PubMed.³ Medical literature often appear as unstructured text in scientific publications, so that clinicians need to carefully read

through a variety of publications to master the latest medical research results on a certain topic. To promote EBM availability in clinical practice, a key cornerstone is the formalization of evidence, making evidence understandable and processable by machines. The PICO framework as stated in [2] is the commonly used representative in EBM, and most pieces of evidence can be characterized by four PICO elements: **Population/Problem** (element P), **Intervention** (element I), **Comparison** (element C), and **Outcome** (element O). Automatic extraction and summarization of medical evidence containing well-articulated PICO elements are widely believed to be the key to easing the burden of clinicians and vastly improving the efficacy of EBM.

In the state-of-the-art research on evidence summarization [3]–[5], this problem was usually solved by supervised classification on medical text. Given a medical publication, an evidence summary is a set of classified PICO sentences. Most of the related work exploited the Cochrane Database of Systematic Reviews⁴ (CDSR) or semi-structured medical abstracts to derive a supervised training of a PICO classification model.

In this work, we present the Aceso system, which learns to automatically produce concise and informative evidence summaries from full-text medical literature. To train Aceso, ~5,000 PICO-related sentences are manually selected and annotated in the full-text of ~3,000 publications. Given a medical text, Aceso comprehensively utilizes medical knowledge and textual features to accurately curate evidential information and organizes these information into a human-friendly summary. Our contributions in Aceso system are as follows:

1) Human annotation is crucially important and meanwhile onerous and expensive. We adopt the active learning paradigm to minimize human effort in annotation. We publish our annotations for the reproduction and extension of our work.

2) Different with related studies that focus on textual features, we propose an UMLS2Vec model to learn a distributed representation of medical knowledge in UMLS, in order to exploit medical knowledge in evidence summarization.

3) Besides the accuracy of PICO classification, we also consider the summary quality in terms of informativeness and simplicity, which are crucial for human understanding.

In following sections, we will introduce our method proposed in the Aceso system in details, and we will verify its effectiveness through experiments.

Manuscript received October 17, 2019; revised February 23, 2020; accepted March 27, 2020. Date of publication April 3, 2020; date of current version September 3, 2020. This work was supported by the National Key R&D Program of China (2019YFB2101802). It was also supported by the National Natural Science Foundation of China (U1736204), and the National Key R&D Program of China (2017YFB1002801, 2018YFC0830201). (Corresponding author: Xiang Zhang.)

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Digital Object Identifier 10.1109/JBHI.2020.2984704

¹The name of Aceso is rooted from the Greek goddess of the healing process: [Online]. Available: <https://en.wikipedia.org/wiki/Aceso>

²UMLS: [Online]. Available: https://www.nlm.nih.gov/research/umls/about_umls.html

³PubMed: [Online]. Available: <https://www.ncbi.nlm.nih.gov/pubmed/>

⁴CDSR: [Online]. Available: <https://www.cochranelibrary.com/cdsr/about-cdsr>

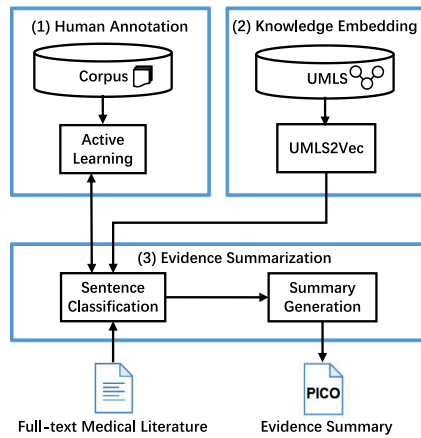


Fig. 1. The system architecture of Aceso.

TABLE I
AN EVIDENCE SUMMARY WITH 3 SENTENCES

Element	Summary
P	"In all 200 patients, the median age was 62 years, and 79 (39.5%) patients were female..."
I/C	"We randomly assigned these patients to receive clopidogrel (75 or 50 mg) plus aspirin (100 mg) or aspirin (100 mg) once daily through 90 days ..."
O	"our findings indicated that the use of 50 mg clopidogrel plus aspirin, and 75 mg clopidogrel plus aspirin was superior to aspirin alone in the aspect of reducing the incidence of ischemic stroke for patients with ICVD..."

II. METHOD

Aceso takes full-text medical literature as input and it outputs a PICO summary of the evidence. Fig. 1 shows the system architecture. Firstly, we collect a large unlabeled corpus for human annotation. Active learning paradigm is adopted to minimize the labeling effort. Then the knowledge embedding module, UMLS2Vec, transforms concepts in UMLS into embeddings. In the third step of evidence summarization, human annotations and concept embeddings are used for the training of a sentence classification model. Concise and informative evidence summaries are finally generated by minimizing the information redundancy in summaries.

Each sentence in a given clinical publication is classified into four labels: P, I/C, O and N. As a common practice stated in [3], [4], we combine the element C with the element I, because a "comparison" usually concerns with one or more "intervention" in an randomized controlled trial (RCT). These two elements are usually intertwined and indistinguishable at the sentence level. The label N stands for sentences that are not relevant to the evidence. In Table I, for the space limit, we present an abridged version of the Aceso-generated summary on the evidence published in [6] as an example, which contains three sentences to help understand the evidence of the dual antiplatelet therapy in Ischemic Cerebrovascular Disease.

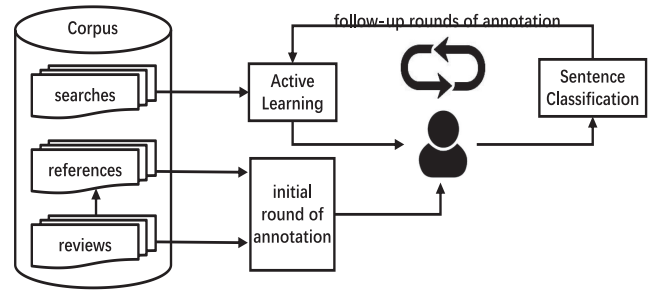


Fig. 2. The process of multi-round human annotation. "reviews" are systematic reviews selected from CDSR. They include human-synthesized evidence summaries; "references" are medical literature, which have been referred and summarized by "reviews"; "searches" represent a large amount of literature searched in PubMed. They are the corpus for active learning.

A. Human Annotation and Active Learning

A process of multi-round human annotation using active learning is shown in Fig. 2. With the help of active learning, selected sentences in the corpus are manually labelled into one of the four elements.

In the initial round of annotation, human annotators are provided with systematic reviews selected from CDSR ("reviews" corpus in Fig. 2). Human-synthesized summaries in "reviews" help annotators quickly and precisely locate PICO sentences in the original text of referred literature ("references" corpus). A small amount of high-quality annotations are exploited for the startup of model training. Next, multiple follow-up rounds of annotation will be continuously driven by active learning, which automatically selects a number of distinct and informative new unlabeled sentences searched from PubMed ("searches" corpus in Fig. 2) and hands out these unlabeled sentences to human annotators. As stated in [7], the aim of active learning is to help the model achieve as great accuracy as possible with limited amount of training data. In our scenario, the potential contribution of an unlabeled sentence is its usefulness that can help to improve the performance of sentence classification. The usefulness of each unlabeled sentence is determined by the following three factors:

a) **uncertainty**: The unlabeled sentence with the highest uncertainty is the one that is least certain to be classified, which makes it highly valuable to be manually labeled by experts. We use the entropy of an unlabeled sentence as a measurement of uncertainty, as shown in Eq (1). Given an unlabeled sentence x , its entropy $u(x)$ is determined by the class posteriors of x on each possible sentence label y . $P_c(y|x)$ is the class probability estimates by a sentence classifier.

$$u(x) = - \sum_{y \in \{ "P", "I/C", "O", "N" \}} P_c(y|x) \log P_c(y|x) \quad (1)$$

b) **diversity**: Redundancy should be avoided in the selection of unlabeled sentences. As shown in Eq(2), the diversity of an unlabeled sentence $d(x)$ is the minimum distance between it and every other sentence x' that is already labeled (denoted by L)

or just been selected for labeling (denoted by Q). $dist(x, x')$ is a linguistic dissimilarity metric, which in our work is Euclidean distance.

$$d(x) = \min_{x' \in LUQ} dist(x, x') \quad (2)$$

c) representativeness: To avoid the misleading outliers, the representativeness of an unlabeled sentence is also an indispensable factor in sentence selection. An unlabeled sentence similar to many other unlabeled sentences is a representative sample and it is highly valuable to the training of the model. Shown in Eq(3), N_k^x is the set of top- k unlabeled sentences that are linguistically similar to x . The representativeness of x is the average distance to its k -nearest unlabeled neighbors.

$$r(x) = \frac{1}{k} \sum_{x' \in N_k^x} 1 - dist(x, x') \quad (3)$$

The overall usefulness of x is a linear combination of all the three factors, which is shown in Eq(4) and Eq(5). w represents the proportion of each factor. Before each round of labeling, a batch of most useful unlabeled sentences will be selected and handed out to human annotators.

$$useful(x) = w_u \cdot u(x) + w_d \cdot d(x) + w_r \cdot r(x) \quad (4)$$

$$\sum_{i=u|d|r} w_i = 1 \quad (5)$$

B. Representation Learning of Medical Knowledge

After human annotation, a routine pipeline of open-domain text summarization task is usually a sequence of sentence classification and sentence summarization. But evidence summarization is special in that: it is not only text-driven, but also driven by intensive domain knowledge. Practically, medical knowledge plays a crucial role in identifying clinical evidence. This motivates us to fully exploit the deep structure in medical knowledge, and transform them into computable knowledge features to support sentence classification.

UMLS is an integration of comprehensive biomedical concepts and terms, presenting their meanings and hierarchical categories. To represent medical knowledge in UMLS, we propose a network embedding model named UMLS2Vec, which exploits the UMLS semantic network. Vector representation of each concept is learned by capturing its semantic type and relations described in the semantic network. Comparing to other types of network embedding models, path-based models such as DeepWalk [8] and Node2Vec [9] show merit on scalability in a wide range of applications. However, there are two issues in applying path-based models for concept embedding: 1) UMLS semantic network is a heterogeneous network, which contains 127 semantic types and 54 relationships. However, most of the existing path-based models only deal with homogeneous networks. 2) There are two kinds of similarities between a pair of concepts in UMLS: the homophily similarity and the structural equivalence. For homophily, concepts that are highly interconnected should be embedded in closely-related representations, such as “heart disease” and “heart”. But structural equivalence

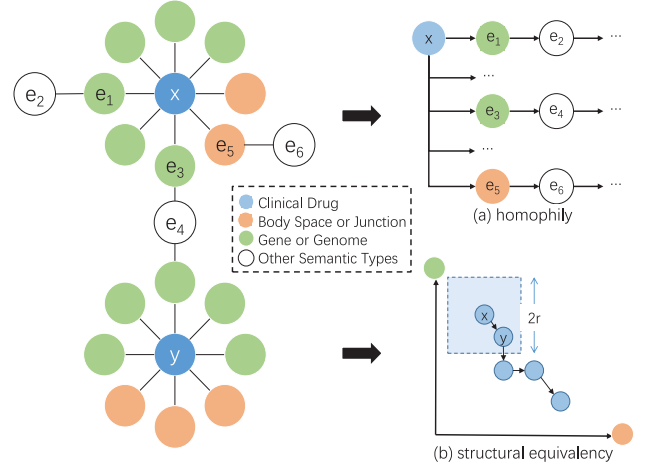


Fig. 3. An example of path finding under the strategy of: (a) H-path and (b) S-path.

does not emphasize connectivity. Two far-away concepts can still belong to a similar structural role, such as two targeted drugs both identifying certain biological targets in a certain living organism.

While most of the existing models merely focus on the homophily similarity, UMLS2Vec model clearly encodes two types of conceptual similarity. Given a concept x , it is used as a starting point to generate a certain number of fixed-length paths under two different strategies: 1) homophily-oriented path-finding strategy (H-path for short); 2) structural-equivalency-oriented path-finding strategy (S-path for short).

The input of the UMLS2Vec is the semantic network of UMLS $g = \langle V, E, C \rangle$, where V represents the set of concepts in UMLS; E represents relationships between concepts; C is the set of semantic types of concepts. As shown in Fig. 3, the nodes, such as x , y , and e_i , are connected concepts in UMLS. Colors of nodes encode the semantic type of corresponding medical concepts. The model of UMLS2Vec transforms the semantic network into a sampled set of concept paths, which are random-walk-paths (denoted as P) generated by H-path (denoted as P^h) and S-path (denoted as P^s).

In H-path strategy, as shown in Fig. 3(a), a random walk from concept x is denoted as $W_x = \langle x, W_x^1, W_x^2, \dots, W_x^l \rangle$, where l is the length of the path. It is a stochastic process with random variables $W_x^1, W_x^2, \dots, W_x^i$, such that W_x^{i+1} is a concept randomly chosen from the direct neighbors of the predecessor W_x^i . For each concept in UMLS, a number of random walk sequences are generated and put into P^h .

In S-path strategy, two concepts share similar patterns if they tend to connect to the same types of neighboring concepts. Given a concept v_i in UMLS, we define $\vec{v}_i = [v_{i1}, v_{i2}, \dots, v_{i|C|}]$ as a $|C|$ -dimensional vector showing the neighboring distribution of v_i over all semantic types. In $\vec{v}_i$, each component v_{ik} represents the number of v_i 's direct neighbors that belong to a certain semantic type $c_k \in C$. Two concepts v_i and v_j are structurally similar if $\vec{v}_i$ and $\vec{v}_j$ are close. Given a starting concept v_i in the $|C|$ -dimensional space, a virtual hypercube taking v_i as

the center can be defined, which has an edge length of $2r$ and is used for outlining the scope of structurally-close neighbors of v_i . Shown in Fig. 3(b), the two-dimensional vectors (for “Clinical Drugs,” there are two neighboring types: “body space and junction” and “gene or genome”) representing the concepts x and y are (2,6) and (3,5). Taking x as the starting concept, the shaded square in Fig. 3(b) outlines the neighborhood of x . A random neighbor in the square will be selected as the successor of x in S-path. Iteration continues until a fix-length concept sequence is generated and put into P^s .

After the generation of H-path and S-path, the skip-gram model [10] is used to work on the concept sequences to produce concept embedding. Here we omit the details of the skip-gram training on concept sequences.

Medical literature can be modeled as a collection of sentences $S = \{s_1, s_2, \dots, s_n\}$, where $s_i = \langle w_i^1, w_i^2, \dots, w_i^m \rangle$ is a medical sentence comprising a list of words. With the help of MetaMap [11], each word in a sentence will be mapped into the UMLS conceptual space: $s_i \rightarrow u_i$, where $u_i = \langle c_i^1, c_i^2, \dots, c_i^m \rangle$, and each c_i^* is a corresponding CUI (concept unique identifier) in UMLS for a given w_i^* . For a non-medical word that cannot be mapped to an UMLS concept, its CUI is defined as Φ , which means a NULL concept. For numeric quantities, MetaMap usually recognize them as “Quantitative Concept” or NULL depending on their contexts. Both linguistic features and knowledge features of the sentence will be exploited in sentence embedding. The distributed representation of a word is a concatenation of its word vector and its concept vector. Each sentence is a feature matrix, in which each row vector represents a word.

C. Evidence Summarization

In Aceso, the Sentence Classification classifies each sentence into one of the four PICO elements. The Summary Generation produces a summary for the evidence in interest and makes sure the summary is concise and informative to the audience.

In Sentence Classification, we implement two classification models: CNN (Convolutional Neural Network) and Bi-LSTM (Bi-directional Long Short-Term Memory), and compare them in their performance on classifying PICO sentences. The CNN model [12] for short-text classification has proven to be effective in many NLP tasks. Bi-LSTM model [4] is a recurrent neural network commonly used in long-text classification.

To generate concise and informative summaries, we propose a two-step approach to eliminating redundancy in evidence summaries: 1) we measure the salience of each PICO sentence by TextRank [13] and abandon trivial sentences; 2) we calculate inter-sentence information redundancy and generate a summary with minimized overall redundancy.

TextRank is a PageRank-style text summarization algorithm. We use TextRank to identify top-20 salient sentences that carry PICO information. The salience of each sentence is measured by its structural importance in a sentence network forming by word co-occurrence. A sentence with high salience implies that it conveys critical PICO information about the evidence. After

TABLE II
STATISTICS OF THE HEART DISEASE DATASET

Heart Disease Dataset			
#reviews	39	#P	398
#references	261	#IC	828
#searches	3,000	#O	803
#sentence	457,000	#N	2,392

the removal of trivial sentences, a set of salient sentences $S = \{S_p, S_{ic}, S_o\}$ is kept as candidates.

To minimize the information redundancy, we define an asymmetric measurement of information coverage between two sentences in Eq(6). s_i and s_j are two sentences in S . $token(s_i)$ stands for the set of words in s_i . $cov(s_i, s_j)$ indicates how much information carried in s_j is also carried in s_i , in another word, in what degree s_j is replaceable by s_i in the final summary. Given a length limitation of k sentences, an optimal combination of k non-redundant sentences S^* are selected from S , as the objective function defined in Eq (7).

$$cov(s_i, s_j) = \frac{|token(s_i) \cap token(s_j)|}{|token(s_j)|} \quad (6)$$

$$S^* = \arg \min_k \sum_{s_i, s_j \in S} cov(s_i, s_j) \quad (7)$$

III. EVALUATION

In this section, we present the statistics of our corpus and datasets, an evaluation on the effectiveness of each step and the overall quality of summarization.

A. Dataset and Experimental Setting

To evaluate our approach, we construct a dataset on heart disease literature from PubMed. Table II shows the basic statistics of the dataset, in which we present the number of original publications and the number of labeled sentences. We use a 7:3 train/test split on labeled sentences in the following experiments. Besides, we create a set of 15 evidence summaries that are manually generated by 3 human experts. These ground truths provide an open platform for benchmarking the quality of evidence summarization.

Hyper-parameters of DeepWalk and word2vec follows the optimal setting given in their original papers [8], [14]. Other parameters in our implementations are optimized via a process of grid search. The source code of Aceso, model parameters, together with the dataset are published in our GitHub project.⁵

B. Evaluation of Active Learning

To evaluate the effectiveness of active learning on reducing human effort of labeling, we compare the performance of two different paradigm of sampling: active learning (AL in short) vs. random sampling (RS) in each round of sentence selection. In RS, unlabeled sentences will be randomly selected for human

⁵ Aceso GitHub: [Online]. Available: <https://github.com/wds-seu/Aceso/>

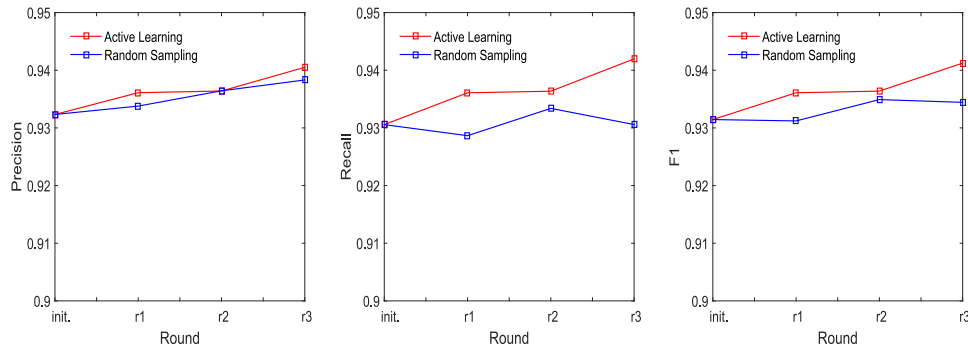


Fig. 4. Average of Precision, Recall, and F1 using active learning or random sampling.

TABLE III
STATISTICS OF EACH ROUND LABELING

label	init.	r1		r2		r3		#total
		AL	RS	AL	RS	AL	RS	
#P	636	0	25	3	9	6	19	398
#IC	760	0	35	5	7	15	6	828
#O	653	9	41	13	26	14	47	803
#N	650	391	299	379	359	365	328	2,392
#total	2,699	400	400	400	400	400	400	5,099

“init.” is the initial round of labeling on references; r_1 - r_3 stand for three rounds of labeling by experts; “AL” is our approach of active learning; “RS” is the random sampling as a baseline.

labeling. Table III elaborates on how much sentences are labeled by a four-round AL-driven or RS-driven human labeling. In each round of labeling, 400 new unlabeled sentences will be automatically selected.

For the weighting scheme among uncertainty, density, and diversity in AL, we reuse the setting of 1:1:1 stated in [15]. In this research, the performance of AL under different weighting scheme was analyzed based on over ten datasets in various fields. It turned out that 1:1:1 is the optimal setting.

After each round of AL or RS labeling, manually-labeled sentences will be used for the PICO classification. In this experiment, we use CNN as the classification model. The macro average of Precision/Recall/F1 after 10 times of running is shown in Fig. 4. With the same amount of manual labeling, RS shows a state of instability and cannot effectively enhance the model, while AL is quite stable in improving the model performance. Another interesting observation is: when sampling sentences with different PICO elements, there is a significant data imbalance, which has been clearly shown in Table III. Especially in AL, few PICO-related sentences (#P/#IC/#O) have been sampled for annotation comparing to the number of non-related sentences (#N). However, those non-related sentences still enhance the model performance, since these AL-selected irrelevant sentences are the most dissimilar ones to the labeled sentences, and these sentences provide diverse features for our model to distinguishing PICO sentence from non-PICO ones. This experiment shows the effectiveness of AL on imbalanced training data.

TABLE IV
PRECISION, RECALL AND F1-SCORE OF SENTENCE CLASSIFICATION WITH DIFFERENT EMBEDDINGS

Model	Embed. *	P	R	F1
CNN	w	0.9281±0.0096	0.9290±0.0104	0.9285±0.0100
	w+d	0.9366±0.0059	0.9370±0.0066	0.9367±0.0062
	w+u	0.9373±0.0046	0.9365±0.0055	0.9369±0.0049
Bi-LSTM	w	0.9234±0.0033	0.9240±0.0034	0.9237±0.0033
	w+d	0.9336±0.0048	0.9288±0.0062	0.9312±0.0054
	w+u	0.9374±0.0044	0.9395±0.0043	0.9384±0.0043

*“w”: only word embedding; “w+d”: a concatenation of word embedding and a DeepWalk-based concept embedding; “w+u”: a concatenation of word embedding and the UMLS2Vec-based concept embedding.

C. Evaluation of UMLS2Vec

In our experiments on UMLS2Vec, the edge length of the hypercube in S-path is set to $5.5e-07$, which leads to a hypercube containing ~ 100 neighbors for each medical concept. To evaluate the effectiveness of UMLS2Vec, a comparative experiment is conducted on the performance of PICO classification using three embedding strategies. Both CNN and Bi-LSTM are adopted as the classification model.

The experimental results are shown in Table IV. It turns out that the concatenation of word embedding and concept embedding (w+d, w+u) is obviously better than using the word embedding (w) alone. It shows the effectiveness to fuse both linguistic and knowledge features in PICO classification. Besides, although very close, UMLS2Vec slightly outperforms Deepwalk on most indicators for both CNN and Bi-LSTM, showing a better capability on capturing knowledge features in UMLS. It is also notable that, comparing to use only linguistic features, knowledge-enhanced classifications have smaller standard deviation after multiple times of running. It indicates that the exploitation of medical knowledge helps to train a stable PICO-classification-model.

D. Evaluation of the Sentence Classification

To evaluate the performance of sentence classification, we conduct two experiments to compare our approach against several state-of-the-art works on two different datasets.

TABLE V
COMPARISON OF ACESO AGAINST OTHER METHODS ON PRECISION, RECALL, AND F1-SCORE

Dataset	Method	P			I/C			O			Macro-F1
		P	R	F1	P	R	F1	P	R	F1	
489K	Jin2018	0.885	0.828	0.856	0.749	0.815	0.781	0.845	0.832	0.838	0.825
	Yuan2019	0.925	0.838	0.879	0.842	0.789	0.814	0.886	0.897	0.891	0.861
	Aceso	0.947	0.867	0.906	0.773	0.850	0.810	0.867	0.918	0.891	0.869
EBM-NLP	Nye2018	0.780	0.660	0.710	0.610	0.700	0.650	0.690	0.580	0.630	0.663
	SciBERT	—	—	—	—	—	—	—	—	—	0.723
	Aceso	0.824	0.583	0.683	0.781	0.733	0.756	0.816	0.808	0.812	0.750

The first experiment is to compare Aceso with studies proposed by Jin *et al.* in [4] and Yuan *et al.* in [5]. These studies both worked on a dataset of 489 k semi-structured medical abstracts extracted from MEDLINE.⁶ Each abstract contained information about seven elements of evidence: participants, intervention, outcome, aim, results, and conclusion. In [4], Jin used a Bi-LSTM model fed with only linguistic features to identify PICO sentences, and in [5], Yuan proposed a classification model based on a soft-margin SVM with a specialized feature engineering method. In this experiment, we train Aceso with these medical abstracts, and we compare our model with theirs only on classifying PICO sentences, and we ignore other elements in medical abstracts. As shown in the first half in Table V, on classifying P-sentence, Aceso outperforms both Jin and Yuan's models. In terms of I/C and O, there is an intense competition between Aceso and Yuan's model. Overall, Aceso shows its merit on Macro-F1.

The second experiment is conducted on the EBM-NLP corpus published in [16]. EBM-NLP contains ~5,000 medical abstracts describing clinical trials. Words or phrases in this dataset have been manually labeled with PICO tags. Using this dataset, Nye *et al.* proposed a Bi-LSTM-CRF model in [16] for PICO classification. Besides, SciBERT is another challenging method proposed by Beltagy *et al.* in [17], which focused on a pre-training language model working on this dataset. We also train Aceso with EBM-NLP dataset. As shown in the second half in Table V, Aceso has the best overall Macro-F1 over other models. This result strengthens our conclusion that fusing knowledge with linguistics can obviously improve the performance of PICO classification.

E. Error Analysis of Classification Results

We present some examples of classification errors in Table VI. The column "Aceso" and "Gold" are the results provided by Aceso and human ground truths. We observe that it is hard for the classifier to distinguish several pairs of labels, which is caused by their shared features. For example, in the second sentence in Table VI, the words "patient" and "age" make the classifier wrongly label this sentence as a P-sentence, while human experts correctly identify the pattern: "we found a ... reduction in..." as an O-sentence. This sentence also raises

TABLE VI
EXAMPLES OF CLASSIFICATION ERRORS

Sentence	Aceso	Gold
"... past medical events were recorded and thereafter all the patients were evaluated for Bishop's score by a senior resident physician."	P	I/C
"We found a statistically significant reduction in the frequency of MACCE in patients aged <65 years."	P	O
"The dual therapy of 75 mg clopidogrel plus aspirin resulted in a worrisome trend in ..."	I/C	O
"Inability to design a double blind clinical trial due to the different route of drug administration was a major limitation..."	I/C	N

TABLE VII
CONFUSION MATRIX FOR CLASSIFICATION ERRORS

	P	I/C	O	N
P	0.930	0.049	0.014	0.008
I/C	0.039	0.922	0.026	0.013
O	0.019	0.035	0.940	0.006
N	0.007	0.030	0.019	0.944

another issue: although it is basically an O-sentence, the phrase "aged < 65 years" does describe information of Population. It motivates us to study multi-label classification or phrase-level fine-grained classification in our future work.

Table VII details the classification errors for each label in the form of a confusion matrix. Rows correspond to predicted labels, and columns correspond to true labels. The data in the matrix is the ratio of the number of predicted labels to the number of true labels. It can be seen from the matrix that P-sentences are prone to be wrongly classified as I/C-sentences. The classification P- of O- sentences has a higher precision than the classification I/C-sentences. According to our analysis, both P- and I/C-sentences contain descriptions of patients and diseases, which may decrease the classification performance.

F. Evaluation of the Summary Quality

In this experiment, we compare our approach of Aceso-full against four summarization methods as baselines: abstract, RobotReviewer3⁷(RR3), Aceso-random, Aceso-concise. The

⁶MEDLINE: [Online]. Available: <https://www.nlm.nih.gov/bsd/medline.html>

⁷RobotReviewer3: [Online]. Available: <https://robotreviewer.vortexsystems/>

TABLE VIII
COMPARISON OF SUMMARY RESULTS USING ROUGE

Method	ROUGE-1	ROUGE-2	R-W-1.2
abstract	0.283	0.095	0.113
RR3	0.436	0.342	0.204
Aceso-random	0.279	0.102	0.124
Aceso-concise	0.379	0.334	0.038
Aceso-full	0.492	0.366	0.211

ROUGE-n is calculated using the number of shared unigrams (n-grams); ROUGE-W-1.2 (R-W-1.2) is calculated using the union of the longest common subsequences and it takes into account the presence of consecutive matches.

method abstract is a simple baseline, which directly uses the abstract of a publication as its summary; RR3 is a challenging state-of-the-art system proposed by Marshall *et al.* in [18], which uses machine learning models to semi-automate biomedical evidence synthesis; Aceso-random and Aceso-concise are two abridged version of Aceso-full. Aceso-random randomly selects PICO sentences to form a summary after PICO classification, ignoring the conciseness and informativeness; Aceso-concise uses TextRank to generate concise summaries, ignoring the informativeness. We use 15 human-generated evidence summaries as ground truths to evaluate the summary quality. Each ground-truth summary contains 10-15 human-selected sentences. ROUGE,⁸ as a set of widely-accepted summary quality measurements, is used to evaluate the agreement between machine-generated summaries with the ground truths. The results are shown in Table VIII.

We observe that the length of the summary has a remarkable impact on the summary quality when ROUGE is used as the quality indicator. In this experiment, the average length of summaries produced by abstract method is 9; RR3 produces summaries with an average sentence length of 8. To make a fair comparison, we also fix the length of Aceso summaries to 8, which is identical to RR3.

The results clearly show that Aceso significantly improve the summary quality. Aceso-full performs greatly on all three ROUGE-based indicators. To interpret these ROUGE scores, we investigate two state-of-the-art works on text summarization in the medical domain. In [19], the result of quality evaluation was: (R-1=0.7886, R-2=0.3529, R-W-1-2=0.1113), and in [20] the result was (R-2=0.3330, R-W-1-2=0.0782). It is not suitable to compare these two works directly with Aceso, because the summaries they generate were not PICO summaries. However, their results on ROUGE scores provide a good reference point to the quality of Aceso summaries.

We also investigate various length settings. If we gradually increase the number of sentences in summaries, the ROUGE of the summaries will also increase, because the PICO information conveyed in summaries become abundant. But if we keep increasing the length, the ROUGE of lengthy summaries will start to decrease due to topic drift and noisy information. By

a grid search on summary length, we find out that the optimal summary produced by Aceso contains around 17 sentences.

IV. RELATED WORKS

Evidence summarization of medical literature based on PICO is effective in assisting clinical medicine. This task can be described into three steps: corpus construction, PICO element detection, and summary generation.

Corpus construction methods for training PICO classification models includes manual annotation [16], [21], pattern recognition [4], supervised distant supervision [3], etc. Many studies, such as [4], [16], [21], build corpus on medical abstracts, while some others like [3] extend the scope to full-text medical literature. Due to the concise nature of medical abstracts, we believe that many important details of evidential information lie in the full text. In another aspect, comparing to accurate but onerous human labeling, supervised distant supervision [3] can quickly generate a large amount of labeled data, but it is difficult to accurately evaluate its effectiveness without the participation of human experts. We propose to use active learning paradigm to effectively and efficiently minimize the amount of manual labeling.

Many machine learning models have been proposed to detect PICO elements, including Naive Bayes [22], [23], Random Forest [22], Support Vector Machines [5], Conditional Random Fields [21] and multilayer perceptions [24]. Deep learning has been widely used in PICO classification in recent years. In [4], Jin used LSTM for PICO detection. SciBERT [17] shows the merit of pre-trained language model on PICO identification. The RobotReviewer3 system [18] uses a rationale-augmented CNN model to discover PICO sentences. However, these models generally focus on internal linguistic features, while ignoring external medical knowledge. Besides, most related studies stop at identifying PICO sentences. But the conciseness and informativeness of the summary are rarely mentioned. We believe these two properties are critical for readable and useful summaries for clinicians.

V. CONCLUSIONS AND FUTURE WORKS

We present the Aceso system, which automatically generates evidence summaries from full-text medical literature. We adopt active learning to minimize the cost of human labeling. We also propose an UMLS2Vec model, which learns vector representation of each concept in UMLS semantic network and fuses medical knowledge with linguistic features in evidence summarization. Experiments show that our approach is better on identifying PICO sentences against previous studies and outperforms baseline methods on producing concise and informative evidence summaries.

In future works, we will study the information extraction on evidence summaries to identify medical entities and relations in evidence summaries. We will also attempt to build a knowledge graph of evidence to promote the browsing, searching, and synthesizing of evidence in medicines, and to provide support for clinical decision-making.

⁸ROUGE: [Online]. Available: <https://pyip.org/project/rouge/>

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